Relative-displacement reformulation of the trajectory predictor, and quantification of the column-6 distribution-shift bias.

Original Model Analysis and Implementation of Objectives 1, 4 and 4-Relative

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Abstract. This report presents the existing Internet of Vehicles (IoV) trajectory-prediction system and two implemented improvements to its downstream Long Short-Term Memory (LSTM) predictor, followed by their combination. The original system combines SUMO-generated vehicle trajectories, simulated V2X missingness, roadside-unit-assisted imputation, and five-second trajectory forecasting. Objective 1 replaces absolute-position prediction with relative-displacement prediction, reducing clean-input ADE from 1.113 m to 0.382 m. Objective 4 measures the training bias caused by near-total unavailability of the follower-speed feature and retrains the LSTM under the measured missingness rate. In the dual-channel evaluation, the resulting ADE is reduced from 31.370 m to 1.564 m. Objective 4-Relative repeats the masked-training experiment with the relative-displacement predictor and a deployable reconstruction anchor, reducing dual-channel ADE from 2.305 m to 0.642 m. All objectives produce trained predictors, comparative evaluations, and quantitative ADE/FDE deliverables.

1 Problem Statement

Connected and Autonomous Vehicles (CAVs) can exchange information about their motion using Vehicle-to-Everything (V2X) communication. In mixed traffic, however, CAVs coexist with Human-Driven Vehicles (HDVs), which do not broadcast the same state information. A trajectory predictor may therefore receive the ego-vehicle history while missing information about nearby vehicles. Communication outages can also remove consecutive observations from the ego vehicle's history.

The original project identifies two forms of missing data:

- **Structural Feature Missingness (SFM):** Neighbor information, such as leader speed, leader gap, or follower speed, is unavailable because a surrounding HDV does not transmit it over V2X. A feature may be absent throughout the observed history.
- **Temporal Cluster Missingness (TCM):** Short bursts of missing ego-vehicle observations occur because of V2X packet losses or communication interruptions.

The central problem is to predict a vehicle's next five seconds of motion from an incomplete three-second observation history. The original system addresses the missing inputs through roadside sensing and temporal imputation. Our work investigates two further questions: whether the downstream predictor can learn a better trajectory representation, and whether training it on ideal follower-speed information introduces error when that feature is mostly absent during evaluation.

2 Original Model and Working Pipeline

2.1 Simulation and input data

The project uses the Simulation of Urban MObility (SUMO) traffic simulator and the TraCI Python interface to record vehicle trajectories. Its highway simulation contains three lanes

and is sampled at 10 Hz. The experiment includes three CAV-penetration scenario folders (10%, 20%, and 30%), each with 20 simulation seeds.

For each prediction sample, the model uses 30 time steps (3 s) of history and estimates 50 future positions (5 s). The stored arrays are

$$X \in \mathbb{R}^{N\times 30\times 7}, \qquad Y \in \mathbb{R}^{N\times 50\times 2}, \tag{1}$$

where N is the number of trajectory windows. The seven input features are listed in Table 1.

Table 1: Input features used by the original LSTM predictor.

Column	Feature	Role
0	Ego speed	Ego-vehicle motion
1	Ego acceleration	Ego-vehicle motion
2	Ego x position	Ego-vehicle position
3	Ego y position	Ego-vehicle position
4	Gap to leader	Neighbor interaction
5	Leader speed	Neighbor interaction
6	Follower speed	Neighbor interaction

2.2 Corruption and dual-channel imputation

SUMO provides complete ground-truth vehicle states, saved as `X_gt`. The pipeline introduces missing entries to create corrupted histories and then reconstructs the affected features before they are passed to the trajectory predictor.

1. **SFM injection:** Neighbor-state features in columns 4–6 are masked according to the surrounding vehicles' communication availability.
2. **TCM injection:** Burst-type missing intervals are introduced into ego-state features in columns 0–3.
3. **Channel 1 (RSU + Kalman filter):** Noisy roadside-unit observations are used to estimate neighboring-vehicle states and reconstruct leader-related information. The implementation also uses fallback handling for follower speed.
4. **Channel 2 (GRU-D):** A temporal recurrent model reconstructs missing ego-state observations using the available sequence information and physical constraints.

After imputation, the reconstructed 30-by-7 history is supplied to a downstream LSTM. The original predictor contains two LSTM layers with 128 hidden units and a fully connected output layer, producing 50 future (x, y) coordinate pairs. It is trained with mean squared error (MSE) using clean `X_gt` histories and corresponding future-trajectory labels.

2.3 Evaluation metrics and original results

The predicted trajectory is compared with the actual future positions using Average Displacement Error (ADE) and Final Displacement Error (FDE):

$$\mathrm{ADE} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{T}\sum_{t=1}^{T}\|\hat{p}_{i,t} - p_{i,t}\|_2, \tag{2}$$

$$\mathrm{FDE} = \frac{1}{N}\sum_{i=1}^{N}\|\hat{p}_{i,T} - p_{i,T}\|_2, \tag{3}$$

Table 2: Original E1 results at the 20% penetration scenario.

Evaluation condition	ADE (m)	FDE (m)
Zero-fill (no imputation)	67.845	73.972
Forward-fill	43.518	45.300
Mean-fill (local)	44.113	46.197
Linear interpolation	43.465	45.176
Spline interpolation	43.429	45.110
Channel 1 only	53.803	56.767
Dual-channel imputation	**31.370**	**33.015**

where $T = 50$. Both errors are reported in meters; lower values indicate more accurate predicted trajectories.

The original dual-channel pipeline obtained the lowest ADE and FDE among these recorded E1 conditions. The baseline absolute-coordinate LSTM also achieved 1.113 m ADE and 1.599 m FDE when tested separately on the clean ground-truth histories used for the Objective 1 comparison. Together, these results motivate examining the trajectory prediction model in addition to the missing-data reconstruction stage.

3 Assigned Work and Motivation

The assignment focused on improving and evaluating the downstream LSTM while retaining the original SUMO dataset and imputation pipeline as the experimental foundation. We implemented two separate objectives, each investigating a distinct source of prediction error, and then combined them in an Objective 4-Relative extension.

Table 3: Purpose of the selected objectives.

Objective	Assigned task	Why it was needed
1: Relative displacement	Train an LSTM to predict relative motion, reconstruct absolute trajectories, and compare against the original model.	Absolute coordinates depend on global location; predicting motion between adjacent steps may be easier for the network to learn.
4: Column-6 bias	Measure follower-speed absence, retrain an LSTM with that feature masked at the observed rate, and compute ADE/FDE differences.	The original LSTM was trained with complete follower-speed information, while evaluation inputs often lack that feature.
4-Relative: Column-6 bias with relative displacement	Repeat the masked-training comparison with the Objective 1 relative-displacement predictor, using a reconstruction anchor taken from the observed input.	The Objective 4 bias was measured for an absolute-coordinate predictor; the relative predictor needs its own measurement, and its anchor must not use ground truth.

Objective 1 changes what the model predicts. Objective 4 changes what information the model sees during training. Objective 4-Relative applies the Objective 4 masking to

the Objective 1 predictor. All use the original two-layer LSTM architecture to keep the investigations focused on these modifications.

4 Objective 1: Relative-Displacement Prediction

4.1 Methodology

The original LSTM directly predicts absolute future coordinates $p_t = (x_t, y_t)$. Objective 1 instead predicts the step-wise displacement between successive positions:

$$\Delta p_t = p_t - p_{t-1}, \qquad t = 1, \ldots, 50, \tag{4}$$

where p_0 is the final observed ego position. During inference, the 50 predicted displacements are accumulated to recover absolute coordinates:

$$\hat{p}_t = p_0 + \sum_{k=1}^{t} \Delta\hat{p}_k. \tag{5}$$

The reconstructed trajectories are evaluated using the same ADE and FDE definitions as the original model.

4.2 Implementation steps

1. Loaded clean trajectory histories (`X_gt`) and future labels (`Y`) from the 10%, 20%, and 30% penetration datasets.
2. Preserved the simulation-seed split: seeds 1–14 for training, 15–16 for validation, and 17–20 for testing.
3. Converted the original position targets into displacement sequences, with normalization parameters computed from training data.
4. Trained the two-layer LSTM with 7 input channels, 128 hidden units, MSE loss, Adam optimizer, learning rate 10^{-3}, batch size 128, and up to 30 epochs.
5. Selected the model using validation loss, reconstructed absolute trajectories from predicted displacements, and compared the saved relative and original absolute models on the same 8,084 held-out test windows.

The best recorded validation MSE was 0.349134 on the normalized displacement targets (epoch 12).

4.3 Results and observations

The relative-displacement formulation reduced ADE by approximately 65.70% and FDE by 41.61% on the shared clean-input test set. The model learned the vehicle's local movement at each prediction step, after which summation recovered its future coordinates. This establishes relative displacement as an effective output representation for the evaluated dataset.

4.4 Objective 1 deliverables

- **Training implementation:** `objective1/run_objective1.py`, including displacement-target construction and trajectory reconstruction.
- **Comparison implementation:** `objective1/compare_baseline.py`, evaluating the original and relative models on the same held-out windows.
- **Trained model:** `objective1/checkpoints/relative_displacement_lstm.pt`.

Table 4: Objective 1: absolute-coordinate versus relative-displacement LSTM on clean test histories.

Metric	Original LSTM	Relative LSTM	Improvement
ADE (m)	1.112899	**0.381688**	**65.70%**
FDE (m)	1.599128	**0.933725**	**41.61%**
Test windows	8,084	8,084	Same test set

- **Result files:** `summary_metrics.csv`, `training_history.csv`, and `baseline_comparison.csv`; a separate `baseline_comparison_per_scenario_seed.csv` provides the per-scenario breakdown.
- **Experiment documentation:** `objective1/config.json`, `logs/run_summary.json`, and `README.md`.

5 Objective 4: Measuring Column-6 Training Bias

5.1 Motivation and measured missingness

Column 6 of the LSTM input contains the follower vehicle's speed. In the original model, the LSTM is trained on complete `X_gt` histories, so this feature is available during training. In mixed HDV/CAV traffic, follower-speed measurements can be structurally absent because HDVs do not transmit V2X information. Objective 4 examines the resulting difference between an LSTM trained with complete inputs and one trained under the observed follower-speed absence rate.

For the 20% penetration training scenario (seeds 1–14), a follower-speed mask bank of shape $(9269, 30)$ was analysed. Its measured absence probability was

$$p_{\text{missing}} = 0.99002769 \quad (99.002769\%). \tag{6}$$

This measured percentage was used as the masking probability for the retrained LSTM.

5.2 Implementation steps

1. Constructed the mask bank from the 20% penetration scenario and calculated the proportion of missing follower-speed entries.
2. Loaded clean `X_gt` training and validation windows across the three penetration scenarios.
3. Independently sampled a missingness decision for every window and time step using the measured probability, and masked only column 6:

$$m_{i,t} \sim \text{Bernoulli}(p_{\text{missing}}), \qquad X'_{i,t,6} = \begin{cases} \text{NaN}, & m_{i,t} = 1, \\ X_{i,t,6}, & m_{i,t} = 0. \end{cases} \tag{7}$$

4. Preserved columns 0–5 and the original future-trajectory labels. Missing follower-speed entries were handled during preprocessing before being passed to the LSTM.
5. Trained a second two-layer, 128-hidden-unit LSTM with the original absolute-coordinate prediction target, using 27,987 training windows and 3,932 validation windows.

6. Evaluated the original clean-trained checkpoint and the new masked-trained checkpoint on matched held-out inputs under zero-fill, spline interpolation, Channel 1, and dual-channel conditions.

The achieved masking percentages were 98.993699% for training and 99.001356% for validation, closely matching the measured absence rate.

5.3 Results and observations

Table 5: Objective 4: original versus column-6-masked-trained LSTM. All errors and differences are in meters.

	ADE (m)			FDE (m)		
Condition	Original	Masked	Reduction	Original	Masked	Reduction
Zero-fill	67.845	39.737	28.108	73.972	45.555	28.417
Spline interpolation	43.490	12.149	31.341	45.132	13.854	31.278
Channel 1 only	53.803	32.490	21.313	56.767	37.482	19.285
Dual-channel	31.370	**1.564**	29.806	33.015	**2.665**	30.350

The difference for each evaluation condition is calculated as

$$\Delta\text{ADE} = \text{ADE}_{\text{original}} - \text{ADE}_{\text{masked}}, \qquad \Delta\text{FDE} = \text{FDE}_{\text{original}} - \text{FDE}_{\text{masked}}. \tag{8}$$

In the dual-channel condition, masked training lowered ADE from 31.370 m to 1.564 m, a reduction of approximately 29.81 m. FDE dropped from 33.015 m to 2.665 m, an improvement of 30.35 m. Improvements were also observed in the other three evaluation conditions. These results quantify how strongly follower-speed availability during training can affect subsequent trajectory prediction when the feature is absent in the evaluated inputs.

5.4 Objective 4 deliverables

- **Measured missingness:** A 20% scenario follower-speed mask bank and empirical absence probability of 99.002769%.
- **Masked-training pipeline:** Preprocessing and retraining procedure that masks column 6 while preserving the other six input channels and original labels.
- **Trained checkpoint:** A separate LSTM trained with realistic follower-speed absence, retaining the original absolute-coordinate target.
- **Matched evaluation:** Original-versus-masked checkpoint ADE/FDE comparison for zero-fill, spline, Channel 1, and dual-channel inputs.
- **Bias measurement:** Per-condition ADE and FDE reductions, including the approximately 30 m improvement in the dual-channel evaluation.

6 Objective 4-Relative: Column-6 Bias for the Relative-Displacement Predictor

6.1 Motivation and design

Objective 4 measured the column-6 bias for the original absolute-coordinate predictor, whereas Objective 1 showed that a relative-displacement predictor is more accurate. Objective 4-Relative joins the two: it asks how much masked training helps when the predictor

outputs relative displacements, and it removes a hidden dependence on ground truth. A relative predictor needs a starting point to turn displacements back into positions, and an earlier version of this experiment used the clean `X_gt` final position for that purpose. A deployed model would not have that value, because the final ego position is itself missing in about 10% of corrupted test windows.

Two models are compared on identical inputs:

- **Original model:** the Objective 1 relative-displacement LSTM, trained with column 6 always present and evaluated without further training.
- **Masked model:** a new, randomly initialised LSTM with the same architecture, trained with column 6 masked at the measured absence rate. The Objective 1 checkpoint supplies only the input and displacement normalisation scalers; its weights are not loaded, so this is training from scratch rather than fine-tuning.

6.2 Implementation steps

1. **Measured the absence rate** from the stored `X_mask` column 6 of the 20% penetration training seeds 1–14, giving $p_{\text{missing}} = 0.99002769$, identical to Objective 4.
2. **Masked training and validation windows.** Every follower-speed entry of clean `X_gt` was hidden independently with that probability across all three scenarios (training mask seed 42, validation mask seed 43). Columns 0–5 and the labels were left unchanged, and the code asserts this. The realised rates were 98.9937% (27,987 training windows) and 99.0014% (3,932 validation windows).
3. **Encoded missing values** as a physical zero before normalisation, with no separate missingness-indicator feature:

$$\tilde{X}_{i,t,6} = \begin{cases} 0, & m_{i,t} = 1, \\ X_{i,t,6}, & m_{i,t} = 0, \end{cases} \qquad Z = \frac{\tilde{X} - \mu_X}{\sigma_X}. \tag{9}$$

4. **Built relative targets** from the clean final position and `Y`, normalised with the Objective 1 displacement scalers (the same formulation as Section 4).
5. **Trained from random initialisation:** two-layer LSTM, 128 hidden units, MSE loss, Adam, learning rate 10^{-3}, batch size 128, up to 30 epochs, seed 42, keeping the lowest-validation checkpoint (best validation MSE 0.349201 at epoch 11).
6. **Evaluated on the same test windows:** 20% penetration, seeds 17–20, 2,685 windows, with column-6 absence of 98.51% in the test mask. Four inputs were scored: zero-fill (`X_corrupted`), spline interpolation, Channel 1 only (`X_ch1_imputed`), and dual-channel (`X_fully_imputed`). The real test column-6 mask was re-applied to every condition after imputation, so both models see the same missing follower speeds.

Predicted displacements are inverse-normalised and accumulated from the anchor, and ADE and FDE are computed against the true future positions as in Section 2.3. The difference between models uses the same definition as Equation (8), $\Delta = \text{original} - \text{masked}$, so a positive value means masked training is better.

6.3 Results and observations

In the dual-channel condition, masked training lowered ADE from 2.305 m to 0.642 m (72.2%) and FDE from 3.759 m to 1.183 m (68.5%). Reductions in the other conditions

Table 6: Objective 4-Relative: original versus masked-trained relative-displacement LSTM (observed anchor, 20% penetration, 2,685 test windows). All errors are in meters.

	ADE (m)			FDE (m)		
Condition	Original	Masked	Reduction	Original	Masked	Reduction
Zero-fill	15.992	5.533	10.459	33.573	10.149	23.424
Spline interpolation	13.448	1.820	11.628	29.059	3.181	25.877
Channel 1 only	6.605	5.422	1.183	11.650	9.876	1.775
Dual-channel	2.305	**0.642**	1.664	3.759	**1.183**	2.576

were 65.4% (zero-fill), 86.5% (spline) and 17.9% (Channel 1 only) for ADE. The masked-trained dual-channel model gives the lowest error of all conditions and of both predictors, including the masked absolute-coordinate model of Table 5 (1.564 m ADE), although the two experiments differ in anchor, input normalisation and test construction, so the comparison is indicative rather than exact. Replacing the clean anchor with the observed anchor changed the reductions little: with the oracle anchor, the ADE reductions were 10.459, 11.408, 1.183 and 1.749 m for the same four conditions.

Table 7: Objective 4-Relative: mean error at selected prediction horizons (ADE at step t, meters). O = original, M = masked-trained.

Condition	Model	0.1 s	0.5 s	1.0 s	2.0 s	3.0 s	4.0 s	5.0 s
Zero-fill	O	1.282	3.187	5.708	11.403	18.345	25.863	33.573
	M	**0.979**	**1.722**	**2.639**	**4.493**	**6.353**	**8.241**	**10.149**
Spline	O	1.155	2.561	4.513	9.182	15.320	22.101	29.059
	M	**0.827**	**0.963**	**1.129**	**1.479**	**1.937**	**2.496**	**3.181**
Channel 1 only	O	1.037	2.007	3.207	5.532	7.725	9.743	11.650
	M	**0.973**	**1.707**	**2.612**	**4.425**	**6.233**	**8.048**	**9.876**
Dual-channel	O	0.381	0.761	1.225	2.052	2.738	3.277	3.759
	M	**0.296**	**0.336**	**0.386**	**0.502**	**0.667**	**0.892**	**1.183**

Table 7 shows how the improvement develops over the five-second horizon. The masked model's advantage grows with the horizon: for dual-channel input it is 22% at 0.1 s and 69% at 5 s, and for spline input 28% and 89%. The 0.1 s errors are dominated by the anchor rather than the model: the observed anchor is on average 0.79 m from the clean final position for the zero-fill, spline and Channel 1 inputs (whose final ego position is missing in 10.17% of windows for zero-fill and Channel 1) and 0.29 m for dual-channel input, which matches the dual-channel 0.1 s error of 0.296 m. For comparison, the clean-input relative model of Objective 1 has a 0.1 s error near 0.011 m.

6.4 Objective 4-Relative deliverables

- **Training and evaluation implementation:** `objective4_relative/objective4_relative.py`, covering mask-rate measurement, masking, scratch training, four-condition evaluation, and anchor handling (`-skip-train` evaluates a saved checkpoint).
- **Trained model:** `objective4_relative/checkpoints/relative_displacement_col6_scratch.pt`.
- **Result files:** `original_matched_summary.csv`, `objective4_summary.csv`, and `objective4_relative_bias.csv` (primary ADE/FDE and reductions); `original_`

`matched_table3.csv` and `objective4_relative_table3.csv` (per-horizon ADE); `objective4_relative_ORACLE_bias.csv` (oracle-anchor reference); `anchor_diagnostics.csv`, `masking_rates_by_scenario_seed.csv`, `training_history.csv`, and `model_comparison.csv`.

- **Run record and documentation:** `logs/run_summary_scratch.json`, `config.json`, and `README.md`.

7 Combined Findings and Conclusion

The three experiments improve different aspects of the downstream trajectory-prediction task. Objective 1 changes the representation of the output trajectory, Objective 4 adapts the LSTM training inputs to the observed availability of follower-speed information, and Objective 4-Relative applies that adaptation to the relative-displacement predictor with an anchor that needs no ground truth. All were implemented using the existing SUMO data and two-layer LSTM architecture.

Table 8: Summary of the completed objectives.

Objective	Implemented change	Main deliverable
1	Predict relative displacements and reconstruct future positions.	ADE reduced by 65.70%; FDE reduced by 41.61% on 8,084 clean test windows.
4	Retrain the LSTM with measured 99.002769% follower-speed missingness.	Dual-channel ADE reduced by about 29.81 m and FDE by about 30.35 m.
4-Relative	Retrain the relative-displacement LSTM with the same missingness and reconstruct from an observed anchor.	Dual-channel ADE reduced from 2.305 m to 0.642 m (72.2%) and FDE from 3.759 m to 1.183 m (68.5%) on 2,685 test windows.

The original model established a complete pipeline from mixed-traffic simulation and missing-data generation to imputation and downstream prediction. Our implementation shows that the downstream LSTM can also be improved by choosing an appropriate target representation and accounting for unavailable neighbor information during training. The deliverables are three independently trained predictors, their experimental implementations, and quantitative ADE/FDE comparisons with the original system.

Project files: IoV Imputation Implementation Plan; original SUMO predictor and E1 results; Objective 1 scripts, checkpoint and CSV results; Objective 4 masking experiment and matched ADE/FDE results; Objective 4-Relative script, checkpoint and CSV results.